Strategy Edge Report — Mean-Reversion (z-score), EUR/USD

Prepared by costwall · 2026-06-18 · Package: Quick Edge Check

What this report is. An honest, costs-included test of whether your strategy has a tradeable edge — and if not, *exactly what stands between it and one*. A red verdict is not a dead end: it means you won't lose your deposit on this as-is, plus a map of what would have to change. We sell the truth and the map, not a stamp.

1. Scope & Assumptions (what we fixed)

Item	Value
Instrument	EUR/USD
Timeframe	H1 (signal lookback 48 bars = ~2 days)
Holding horizon	24 bars = 24 hours
Round-trip cost assumed	3 bps (retail forex realistic)
Validation	causal 50/50 split + rolling walk-forward (18 folds)
Test type	ONE pre-specified strategy → N=1, fair test, NO multiple-testing penalty

Note: timeframe and signal lookback are separate axes; cost is the binding wall for marginal edges (see breakeven below).

2. Verdict

NO_EDGE

Plain English: After costs this mean-reversion rule is a coin flip — and the faint tilt it does show is not statistically distinguishable from luck. The in-sample "profit" disappears out-of-sample. Do not trade it as-is.

Verdict	Meaning
TRADEABLE	Net edge survives costs, is significant, and holds out-of-sample.
SUB_COST	Real predictive signal — but your costs eat it. Fixable (see §4).
INCONCLUSIVE	Not enough independent data yet to decide.
NO_EDGE ←	No directional skill beyond a coin flip. Do not trade this as-is.

3. Evidence

Metric	Value	Reads as
AUC (directional skill)	0.514	≈ coin flip (0.50); below the 0.55 skill line
Net per trade	–1.14 bps	loses after costs
Breakeven cost (`gross_bps`)	**1.86 bps**	lives only if round-trip cost < 1.86 bps

Metric	Value	Reads as
t-stat	–0.35	not significant (need ≥2)
Permutation p (is the signal *real*?)	0.30	**> 0.05 → direction is NOT distinguishable from luck**
Sub-period consistency	0.50	profitable in only half the periods
Max drawdown	323 bps	worst peak-to-trough
Walk-forward	WF_DEGRADED	IS net +3.3 bps → **OOS net –2.9 bps** (flips to loss)
Deflated Sharpe	N/A	single pre-specified strategy — no search to deflate

4. Revival Map — what would make this tradeable

- **Is the signal even real?** Permutation **p = 0.30** → the direction is **not statistically distinguishable from luck**. This is the core stop sign: there is no faint-but-real edge here to rescue, unlike a SUB_COST case.
- **Cost wall.** Breakeven is **1.86 bps** round-trip; you assumed 3 bps. Even a broker under 1.86 bps wouldn't save it — the signal itself isn't significant.
- **Walk-forward tells the real story.** In-sample the threshold finds a +3.3 bps "profit", but out-of-sample it flips to –2.9 bps. That gap **is** the overfit: the in-sample gain was the entry threshold fitting noise, not a repeatable edge.
- **Nearest it came.** On an H4 timeframe AUC rose to ~0.56 — but on a tiny sample (14–30 trades) that does not survive deflation; not a real lead.
- **Data sufficiency.** 113 trades / 182 independent observations — adequate. The verdict is a real "no", not a "not enough data".

Bottom line: don't trade this as-is — you'd be trading an in-sample illusion. We just saved you the drawdown. If you have a *different* idea you believe in, bring it: a real edge passes this same stack (we calibrate on it).

5. Variation Matrix — what else we could test (and what it costs)

Each axis you add is another trial. More variations = more work (price) AND a higher significance bar (we deflate for the search — that's the honest part).

Axis	Tested here	Alternative	What it pulls	+ tests
Timeframe	H1	+ H4, D1	re-scale horizon to real time; each TF a separate run	×2
Holding horizon	24h	sweep 6h / 24h / 72h	2nd degree of freedom → DSR deflation applies	×3
Cost model	flat 3 bps	vol-scaled slippage	breakeven vs slippage in fast moves	×3
Validation	50/50 + walk-forward	— (already included)	—	—
Instrument	EUR/USD	+ GBP/USD, USD/JPY	does the edge generalize or is it one-off?	×2

Pick what's worth it. We'll quote the added work before running anything.

6. Honest disclaimer

Most retail strategies are observed in-sample or cherry-picked, and **most fail an honest out-of-sample, costs-included test** — that's the base rate, not a judgment of your idea. Our value is telling you the truth *before* the market does, and showing you exactly where the line is. Methodology: cost-wall (AUC + net + breakeven), Deflated Sharpe (de Prado), circular-shift permutation, sub-period consistency, rolling walk-forward. Open harness: github.com/costwall/backtest-harness.

---

7. Important — not financial advice

This report is a statistical analysis of historical data — **not financial, investment, or trading advice, and not a recommendation to trade.**

- **A verdict is not a guarantee.** TRADEABLE (or any result) does **not** guarantee future profit. Past statistical behavior does not predict future results; markets change regime, and live execution (slippage, fills, latency, spread, gaps) differs from any backtest.
- **Your decisions, your risk.** All trading decisions, position sizing, risk management, and any resulting profits or losses are **solely your responsibility.**
- **Analysis only.** costwall does not manage funds, place trades, or advise whether you should trade. Never risk capital you cannot afford to lose.

By using this report you agree that costwall accepts **no liability** for any losses or damages arising from your use of, or reliance on, this analysis.

— costwall · honest, costs-included strategy validation · costwall@gmx.de